

KALUGA, Russian Federation

WMO: 277030

Lat: **54.550N**Lon: **36.367E**Elev: **199**StdP: **98.96**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.9	-21.4	-27.8	0.3	-24.5	-24.0	0.4	-21.0	10.2	-4.8	9.3	-3.1	1.3	30	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.5	19.5	27.0	18.7	25.4	18.3	21.1	26.1	20.1	24.8	19.2	23.7	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.3	23.5	18.3	13.5	22.5	17.4	12.8	21.6	62.0	26.5	58.4	24.7	55.4	23.6	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.6	6.7	-28.8	31.0	3.6	1.9	-31.3	32.4	-33.5	33.5	-35.5	34.6	-38.1	36.0		
(5)			-28.8	22.6	3.5	1.1	-31.3	23.4	-33.3	24.0	-35.3	24.6	-37.8	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-7.0	-6.7	-1.9	6.6	12.9	16.1	18.8	16.7	11.4	5.5	-0.7	-5.6	(6)
(7)		DBStd	10.50	6.61	6.82	4.81	4.99	4.59	3.60	2.96	3.10	4.04	4.33	5.43	6.64	(7)
(8)		HDD10.0	2503	528	467	369	126	25	2	0	0	31	151	322	482	(8)
(9)		HDD18.3	4744	786	700	628	351	175	84	31	69	209	399	572	741	(9)
(10)		CDD10.0	889	0	0	0	24	114	185	273	208	73	10	1	0	(10)
(11)		CDD18.3	89	0	0	0	0	6	18	45	19	1	0	0	0	(11)
(12)		CDH23.3	872	0	0	0	4	92	199	369	189	17	0	0	0	(12)
(13)		CDH26.7	164	0	0	0	1	12	36	84	31	1	0	0	0	(13)
(14)	Wind	WSAvg	3.2	3.6	3.6	3.8	3.4	3.1	2.9	2.7	2.6	2.8	3.2	3.5	3.6	(14)
(15)	Precipitation	PrecAvg	660	47	36	39	36	61	78	82	59	67	69	51	44	(15)
(16)		PrecMax	850	72	66	72	63	120	141	151	162	179	188	106	81	(16)
(17)		PrecMin	415	20	10	14	15	15	31	6	3	24	13	3	16	(17)
(18)		PrecStd	113	15	14	18	14	30	34	38	42	40	43	23	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	4.0	12.3	23.6	28.2	29.7	31.2	29.9	25.2	19.1	11.3	7.2	(19)
(20)			MCWB	2.6	2.4	6.6	13.3	17.6	19.4	20.0	20.0	16.3	13.1	9.7	6.3	(20)
(21)		2%	DB	2.5	2.9	8.8	20.5	26.2	27.7	29.0	27.2	23.1	16.0	9.0	4.1	(21)
(22)			MCWB	1.7	1.7	4.2	12.0	17.1	19.1	19.9	19.0	16.1	12.2	7.8	3.2	(22)
(23)		5%	DB	1.3	2.1	6.3	17.9	23.8	25.8	27.1	25.5	20.7	13.8	7.6	2.3	(23)
(24)			MCWB	0.7	0.9	3.0	10.6	15.7	18.2	19.6	18.4	15.3	11.1	6.5	1.6	(24)
(25)		10%	DB	0.6	1.3	4.2	15.4	21.4	23.8	25.3	23.6	18.2	11.8	5.7	1.2	(25)
(26)			MCWB	-0.1	0.5	1.9	9.5	14.4	17.5	19.0	17.5	13.9	9.7	5.0	0.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.0	7.2	14.4	19.3	21.5	22.7	21.9	18.1	14.6	10.3	6.5	(27)
(28)			MADB	3.3	3.5	11.1	21.6	26.2	27.4	28.3	27.1	23.3	17.3	11.3	6.7	(28)
(29)		2%	WB	1.8	2.0	4.9	12.6	17.7	20.1	21.6	20.4	16.7	12.8	8.1	3.4	(29)
(30)			MADB	2.2	2.6	7.9	18.5	23.8	25.5	26.6	25.1	21.2	15.2	8.7	3.9	(30)
(31)		5%	WB	0.8	1.2	3.4	11.5	16.5	19.0	20.5	19.3	15.6	11.3	6.5	1.8	(31)
(32)			MADB	1.2	1.8	5.4	16.8	22.1	24.2	25.1	23.8	19.5	13.4	7.3	2.2	(32)
(33)		10%	WB	0.0	0.6	2.3	10.1	15.4	17.9	19.6	18.1	14.5	10.0	4.9	0.8	(33)
(34)			MADB	0.5	1.2	3.9	14.8	20.3	22.6	23.8	22.5	17.8	11.5	5.6	1.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.2	6.7	7.5	9.9	11.2	10.2	10.8	9.4	6.9	4.6	4.7	(35)	
(36)			MADB	3.7	4.4	9.9	14.2	14.7	13.9	13.1	14.1	13.5	11.1	6.3	3.7	(36)
(37)			MCWBR	3.7	3.9	6.5	7.3	6.8	6.6	5.8	6.7	7.1	7.2	5.4	3.6	(37)
(38)		5% WB	MADB	3.6	4.0	8.3	12.4	12.4	11.8	11.1	12.2	11.3	9.4	5.3	3.6	(38)
(39)			MCWBR	3.7	3.8	5.9	6.9	6.4	6.2	5.7	6.7	6.5	6.5	5.0	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.289	0.328	0.347	0.416	0.393	0.382	0.419	0.430	0.385	0.355	0.327	0.291	(40)	
(41)		taud	2.122	2.052	2.106	2.126	2.302	2.354	2.270	2.226	2.326	2.353	2.239	2.129	(41)	
(42)		Ebn at Noon	627	724	812	803	852	866	821	783	773	708	584	534	(42)	
(43)		Edh at Noon	65	100	119	136	120	116	124	122	97	75	60	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.69	1.59	3.09	4.04	5.41	5.64	5.46	4.53	2.90	1.39	0.55	0.38	(44)	
(45)		RadStd	0.14	0.20	0.36	0.32	0.45	0.56	0.54	0.49	0.32	0.23	0.10	0.11	(45)	